

Three-Stage Hybrid Recommendation System for Personalized Skincare Products and Routines

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A personalized skincare recommendation system presents innovation opportunities for the beauty industry that is capable of both enhancing customer experience and supporting continued market growth. This paper proposes a three-stage architecture for personalized skincare aimed to addressing the challenges of aligning user requirements, ingredient safety, and product quality. The proposed architecture integrates constraint-based filtering, semantic ranking using bi-encoder or cross-encoder, and neural network quality assessment to deliver recommendations for single-product and multi-product (routine) scenarios, thus improving practical applicability of the system in the skincare market. The routine recommendations also incorporate an ingredient compatibility component that ensures safe product combinations while maintaining diversity. The result demonstrates significant performance in the three-stage system that achieves 84.68% requirement satisfaction with 15.8% catalog coverage for single-product recommendations using cross-encoder, and 84.22% satisfaction with 42.72% diversity for routines using bi-encoder. The performance analysis reveals that constraint filtering provides strong foundational accuracy in fulfilling user requirements, while semantic similarity and neural network components enable enhancement in model diversity and quality. This proposed architecture successfully achieve a balance between accuracy, diversity, and computational efficiency that demonstrates its practical viability in the skincare industry.

Keywords: Personalized skincare, Recommendation system, Constraint filtering, Semantic ranking, Neural network